

Threat Intelligence Report

Case study: Advanced Persistent Threat (APT)-Sea Turtle

Investigation of APT Sea Turtle DNS Hijacking & SnappyTCP Malware

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Verdict: Malicious (Confirmed State-Aligned APT Activity)

Analysis Date: April 21, 2026

Table of Contents

1. Executive Summary
2. Overview of Sea Turtle APT
3. Indicators of Compromise (IOCs)
4. Attack Lifecycle & TTPs Analysis (Deep Dive)
 - 4.1. MISP Event Intelligence Validation
 - 4.2. Extracted Indicators & Infrastructure Mapping
 - 4.3. OSINT Verification: VirusTotal & Shodan Analysis
 - 4.4. Operational Infrastructure: VPN & Staging Nodes
 - 4.5. Domain Reconnaissance & Masquerading Analysis
 - 4.6. DNS Hijacking & Initial Access
 - 4.7. Relational Infrastructure Mapping & Redundancy
 - 4.8. MitM Infrastructure & Service Fingerprinting
5. Mitigation & Defence Strategy
 - 5.1. DNS Layer Hardening

5.2. Administrative Portal Security

5.3. Linux Infrastructure Hardening

5.4. Immutable Logging

6. Actionable Detection Logic (Splunk & Wazuh)

6.1. Splunk Detection: Anti-Forensic Execution (T1070)

6.2. Splunk Detection: SnappyTCP Web Shell Activity

(T1505.003) 6.3. Wazuh EDR: File Integrity Monitoring for cPanel

1. Executive Summary

An investigation was conducted on threat intelligence (**MISP Event #447**) concerning Sea Turtle, a highly sophisticated, state-aligned advanced persistent threat (**APT**) operating out of Turkey. Analysis confirms this group is actively executing espionage campaigns targeting critical infrastructure, telecommunications, and IT service providers in the Netherlands. The attack vector employs large-scale DNS hijacking for credential harvesting, followed by the localised deployment of a custom reverse TCP shell (**SnappyTCP**), utilising commodity cloud infrastructure and commercial VPNs to evade detection.

2. Overview of Sea Turtle APT

Sea Turtle, also known as Teal Kurma or Cosmic Wolf, is a sophisticated Advanced Persistent Threat group operating in alignment with Turkish national interests. Their primary mission is long-term espionage, focusing on gathering political and economic intelligence rather than seeking financial gain. This group specifically targets government institutions, telecommunications providers, Internet Service Providers, and NGOs across Europe, the Middle East, and North Africa.

Their operational success relies on a complex infrastructure that includes command-and-control servers, such as **forward.boord.info**, which communicate over **Port 443** to blend into normal encrypted web traffic. To maintain anonymity, they route their activities through commercial VPN services and use deceptive nameservers like

ns1.rootdnservers.com to masquerade as core internet infrastructure.

The group's signature tactic is DNS hijacking, where they manipulate upstream infrastructure at the registrar or registry level to redirect legitimate traffic to attacker-controlled systems. They often perform these redirections for very short durations to harvest credentials while minimising the risk of detection. Once they gain initial access, often by compromising administrative panels like cPanel, they deploy custom malware such as **SnappyTCP**. To ensure persistence, they utilise tools like NoHup to keep processes running and engage in aggressive anti-forensic techniques, including clearing Bash histories and deleting system logs to erase their digital footprint.

Recent campaigns have seen the group evolve to target supply chain vulnerabilities, compromising IT service providers to gain indirect access to multiple high-value victims simultaneously. To defend against these maturing capabilities, organisations must prioritise securing the DNS layer through DNSSEC and Registry Locks, enforcing hardware-based multi-factor authentication on all administrative accounts, and implementing centralised, immutable logging to preserve forensic evidence even if local logs are wiped.

3. Indicators of Compromise (IOCs)

- **Malicious C2 Domains:** forward.boord.info, lo0.systemctl.network
- **Hosting IPs (C2):** 93.115.22.212, 95.179.176.250
- **VPN Nodes (Staging):** 82.102.19.88 (Belgium), 62.115.255.163 (Denmark), 193.34.167.245 (Netherlands)
- **Rogue Nameservers:** ns1.rootdnservers.com, ns1.intersecdns.com •

Malware Payload: SnappyTCP (Reverse Shell)

4. Attack Lifecycle & TTPs Analysis (Deep Dive)

- **Initial Access (DNS Hijacking):** The threat actor bypasses traditional perimeter

defenses by compromising third-party DNS registrars. They alter target Nameserver (NS) records to point to actor-controlled infrastructure, intercepting traffic via Man-in-the-Middle (MitM) nodes to harvest administrative credentials seamlessly.

- **Defense Evasion & Persistence:** Using stolen credentials, the adversary accesses administrative portals (such as cPanel) via VPN nodes located in Europe to mask their origin. They then deploy the **SnappyTCP** malware to establish a persistent backdoor.

4.1. MISP Event Intelligence Validation

Analysis of **MISP Event #447** provided the foundational intelligence for this investigation. The event clusters and **MITRE ATT&CK** tagging confirm the operational scope, victimology, and specific techniques utilised by the adversary.

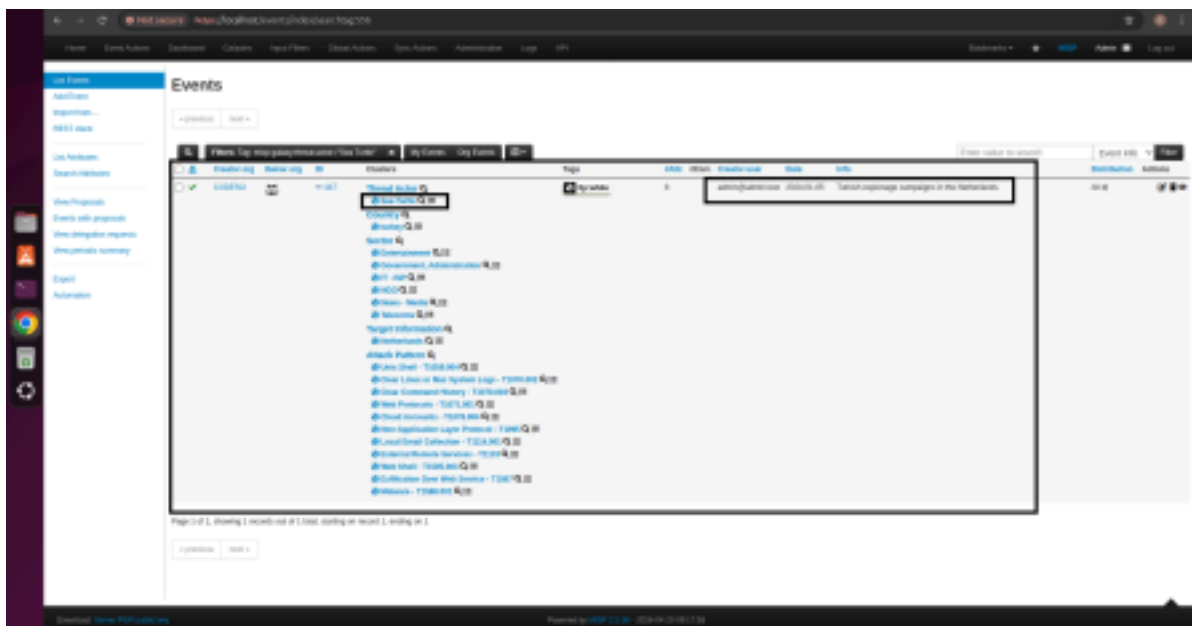


Figure 1: MISP Event #447 – Sea Turtle Campaign Profile

Persistence & Evasion (Anti-Forensics): The event highlights a heavy reliance on Unix shell commands (**T1059.004**) to deploy web shells (**T1505.003**) and malware. Most critically, to evade detection and blind SOC analysts, the threat actor actively executes anti-forensic measures by clearing command histories (**T1070.003**) and completely wiping **Linux/Mac** system logs (**T1070.002**) post-compromise.

Event Overview & Threat Classification

The intelligence is officially documented as **Turkish espionage campaigns in the Netherlands** (Event ID: **447**), providing a verified baseline for our investigation into state-aligned activities.

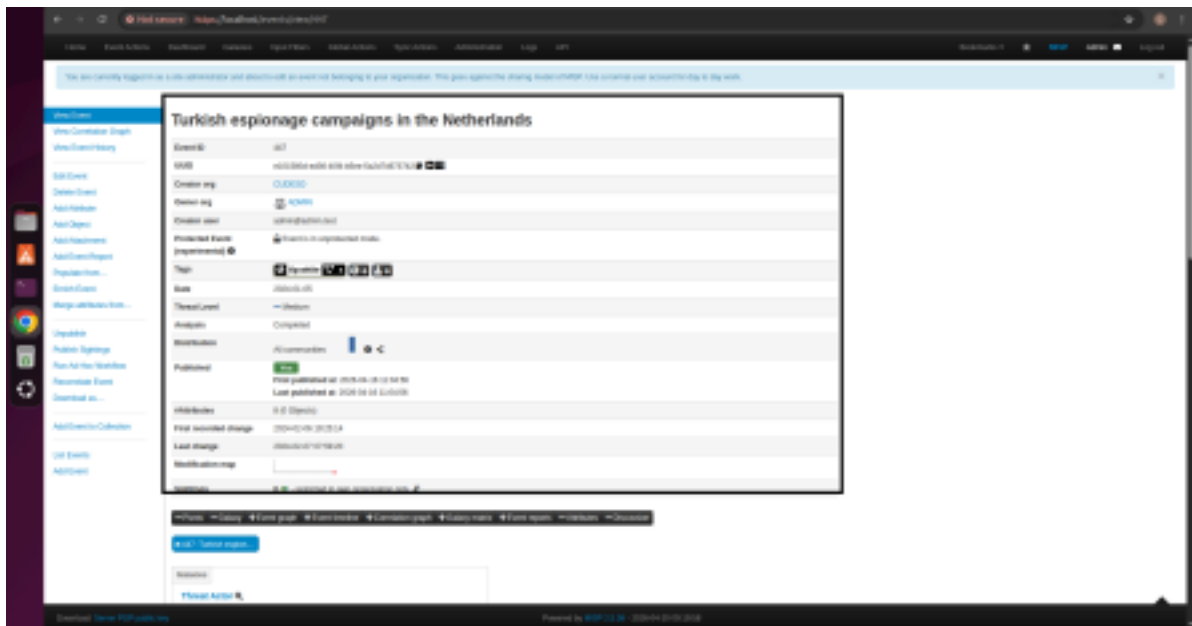


Figure 2: MISP Event #447 Overview – Turkish Espionage Campaigns

The intelligence is classified under **TLP: WHITE**. This is critical for our operational response, as it indicates the intelligence carries no sharing restrictions and can be distributed freely across our internal **SOC**, **GRC** teams, and the tactical board without violating **OPSEC** or **NDA** boundaries. The event is assigned a **medium** threat level with the analysis phase marked as **completed**. This signifies that the intelligence is mature, verified, and ready to be converted into active detection rules.

4.2. Extracted Indicators of Compromise & Infrastructure Mapping

The 8 verified attributes extracted from **MISP Event #447** provide a clear map of Sea Turtle's operational infrastructure. Rather than relying on a single malicious IP, the

adversary utilises a compartmentalised approach, separating their Command and Control (C2) servers from their staging and access nodes.

ID	Date	Category	Type	Value	Tags	Source	Destination	Related Events	Priority	Score	Confidence	Activity	Actions
2024-01-04	2024-01-04	Network activity	Malware activity	forward.boord.info	Malware	93.115.22.212	The malware beaconing IP was used by the attacker to establish a command and control channel with the domain name.	5	High	100	100	Beaconing	View
2024-01-04	2024-01-04	Network activity	Malware activity	lo0.systemctl.network	Malware	95.179.176.250	The IP address is related to a VPN provider in Belgium and was used as a VPN for the attacker to access the victim's network.	5	High	100	100	Beaconing	View
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Figure 3: MISP Event #447 – Extracted Attributes & Infrastructure

Infrastructure Breakdown:

Command & Control (C2) Channels: The threat actor established dedicated C2 channels utilising the custom **SnappyTCP** malware. Network activity beacons out to specific hostnames (**forward.boord.info** and **lo0.systemctl.network**), which currently resolve to backend IPs **93.115.22.212** and **95.179.176.250**.

Operational VPN Nodes (Defense Evasion): To access victim **cPanel** accounts without triggering geographical anomaly alerts, the adversary actively routed their administrative logins through a European commercial VPN provider. This includes nodes in:

- Belgium: 82.102.19.88 (M247 Europe SRL)
- Denmark: 62.115.255.163 (Arelion)

4.3. OSINT Verification: VirusTotal Analysis

To validate the raw intelligence extracted from MISP, independent Open Source Intelligence (OSINT) gathering was conducted against the primary Command and Control (C2) hostname, **forward.boord.info**.

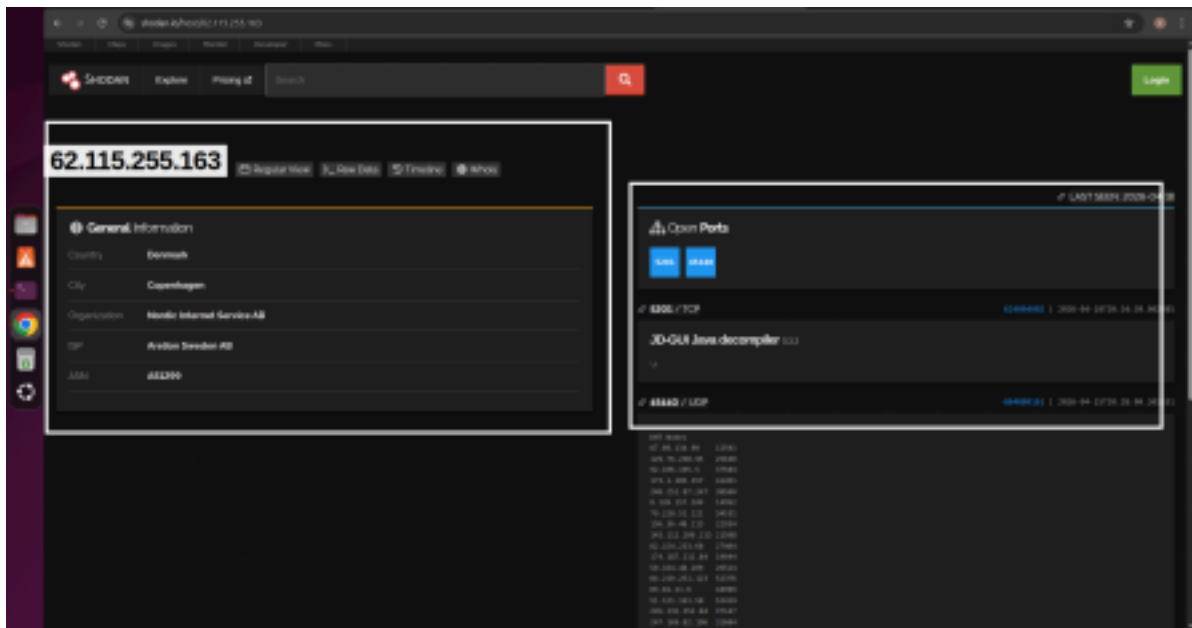


Figure 6: Shodan Infrastructure Analysis – Operational VPN Node 62.115.255.163

Forensic Breakdown:

Intelligence Validation (Arelion VPN): The Shodan scan explicitly corroborates the MISP feed, confirming this IP belongs to **Arelion Sweden AB** (Nordic Internet Service AB) and is geographically located in **Copenhagen, Denmark**. This validates the adversary's **TTP** of routing their administrative cPanel logins through legitimate European commercial ISPs to mask their true origin and bypass geo-fencing controls.

Decentralised Network Obfuscation (Port 45440): Port 45440/UDP reveals a massive list of active Distributed Hash Table (DHT) nodes. DHT is the underlying technology for decentralised Peer-to-Peer (P2P) networks (like BitTorrent). The presence of this P2P traffic indicates the adversary may be utilising decentralised networks for obfuscated file transfers, payload staging, or secondary C2 routing to evade traditional, centralised firewall blocklists.

4.5. Domain Reconnaissance & Masquerading Analysis To investigate the registration and structural integrity of the primary Command and Control (C2) domains, a WHOIS query was executed utilising **MXToolbox** against **lo0.systemctl.network** and **forward.boord.info**.

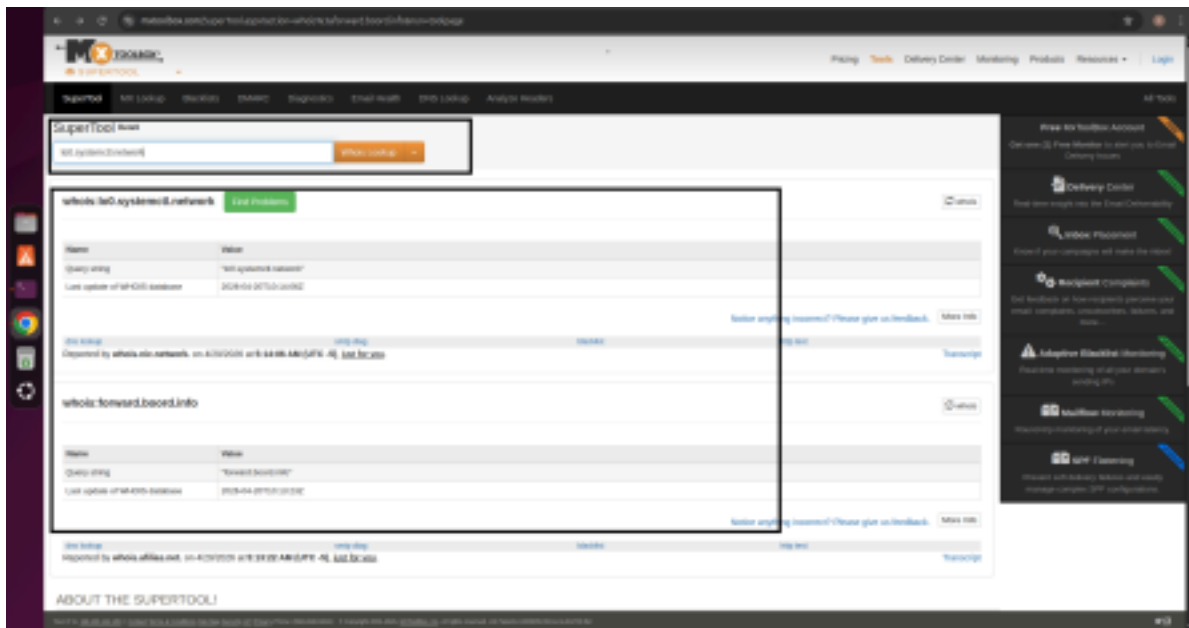


Figure 7: MXToolbox WHOIS Analysis – C2 Domain Reconnaissance

Forensic Breakdown:

- **Defence Evasion (Domain Masquerading):** The most critical finding in this reconnaissance is the structural naming convention of the domain **lo0.systemctl.network**. The adversary is intentionally masquerading their C2 infrastructure to mimic legitimate, internal Linux network traffic.
 - **lo0** mimics the standard loopback network interface.
 - **systemctl** mimics the core systemd service manager in Linux.
- **TLD Exploitation:** The use of **.network** and **.info** Top-Level Domains (TLDs) is highly characteristic of APT C2 infrastructure. These domains are often cheaper and easier to register in bulk, allowing the adversary to spin up and burn infrastructure rapidly as their campaigns evolve.

4.6. DNS Hijacking & Initial Access

Event #198 dates back to 2019 and contains 15 specific attributes. This proves that DNS hijacking is not a new or experimental tactic for this group, but rather a highly refined, core operational capability they have successfully utilised for years.

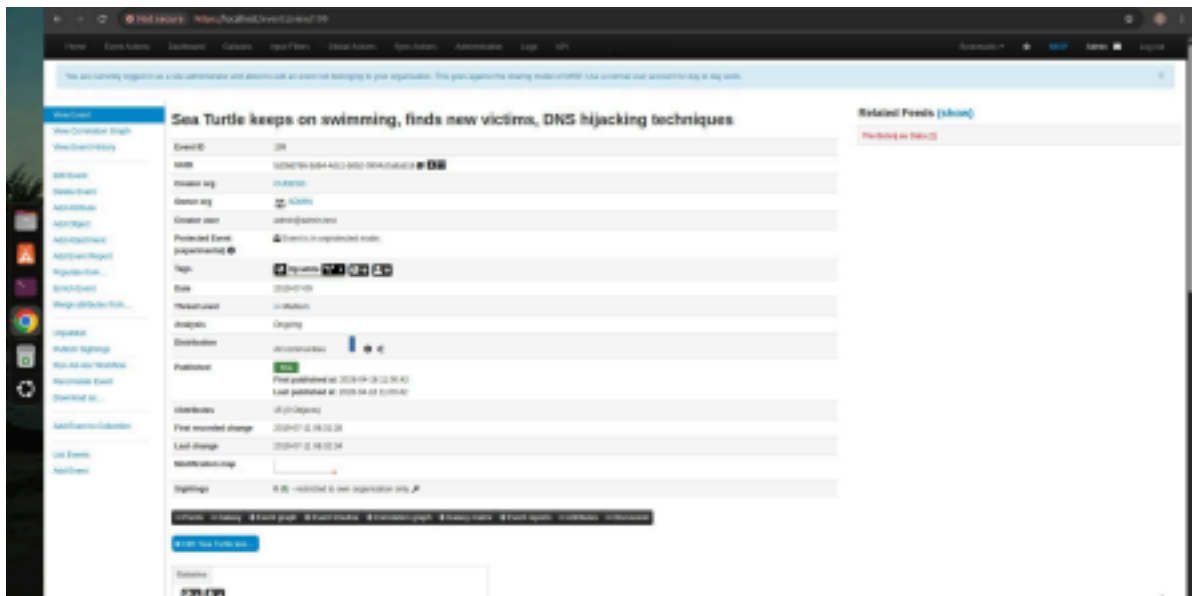


Figure 9: MISP Event #198 Details – DNS Hijacking Campaigns

Similar to our primary intelligence, this historical event is marked **TLP: WHITE**. This allows our SOC to freely utilise these legacy Indicators of Compromise (IOCs) across all internal defensive tools without restriction.

The intelligence is directly correlated with external European threat feeds (**specifically The Botvrij.eu Data**). This cross-referencing ensures our baseline understanding of their DNS manipulation techniques is backed by consensus from international security communities.

DNS Hijacking Infrastructure & Attribute Extraction A deep-dive extraction of the **15** attributes from **MISP Event #198** exposes the specific architecture Sea Turtle utilises to execute their DNS hijacking campaigns. The adversary explicitly relies on domain masquerading and a dedicated Man-in-the-Middle (**MitM**) infrastructure to intercept victim traffic silently.

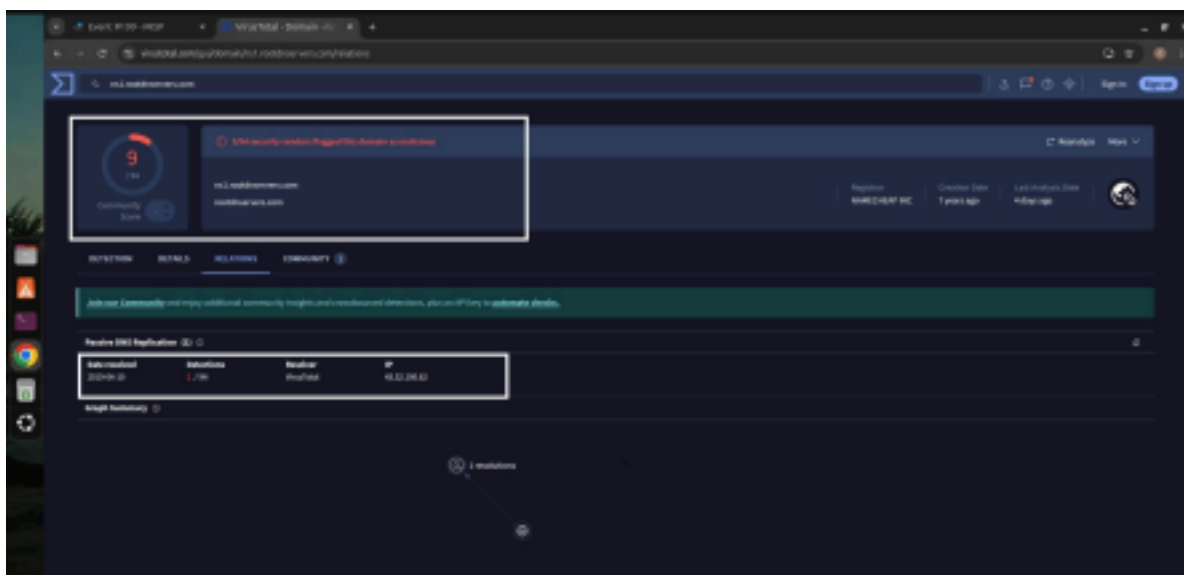


Figure 11: VirusTotal Analysis & Passive DNS – ns1.rootdnservers.com

Threat Vendor Consensus: The domain is actively flagged as malicious by 9 separate security vendors, confirming its status as a recognised threat asset. The domain registration data (Namecheap, created 7 years ago)

Intelligence Validation: This Passive DNS record perfectly matches the Hosted malicious nameserver IP attribute we extracted directly from **MISP Event #198**. By mathematically correlating the domain to the specific IP using external telemetry, this provides high-confidence validation to the board that the provided threat intelligence feed is highly accurate.

Infrastructure Redundancy & Relational Mapping

To fully map the scope of the adversary's DNS hijacking capabilities, extended relational analysis was conducted on the secondary rogue nameserver, ns2.rootdnservers.com, utilising VirusTotal.

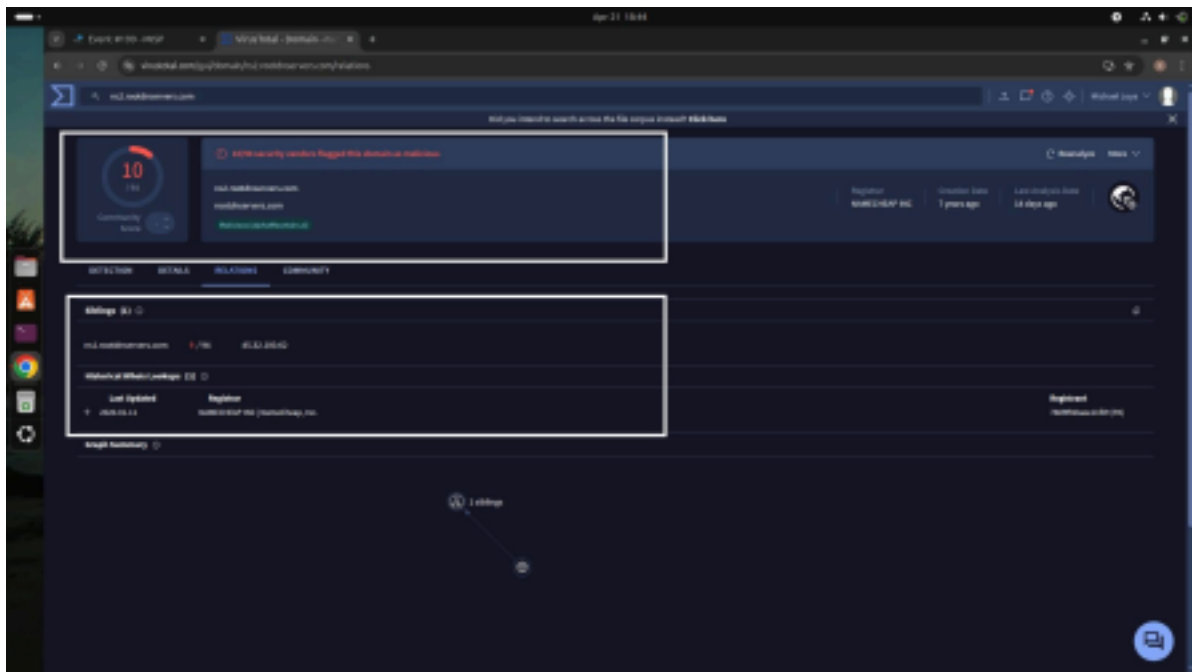


Figure 12: VirusTotal Relational Analysis - ns2.rootdnservers.com

Operational Redundancy: The domain `ns2.rootdnservers.com` is actively flagged by 10 security vendors. By establishing both an `ns1` and `ns2` node, **Sea Turtle** mimics legitimate DNS architecture (which mandates primary and secondary nameservers for fault tolerance). This ensures their Man-in-the-Middle (**MitM**) hijacking campaign remains active even if one node goes offline or gets blocked by defenders.

Sibling Correlation (Infrastructure Clustering): The most critical piece of evidence is the siblings' relational telemetry. VirusTotal algorithmically links this `ns2` domain directly back to our previously analysed `ns1.rootdnservers.com` and its associated **MitM** IP address (`45.32.100.62`). This definitive link proves that these domains do not operate in isolation; they are part of a clustered, centrally managed attack infrastructure operated by the same threat actor.

Infrastructure Diversification & DNS Record Analysis To further validate the breadth of the adversary's hijacking capabilities, a detailed DNS and WHOIS analysis was conducted on the secondary campaign domain, `ns1.intersecdns.com`, using VirusTotal.

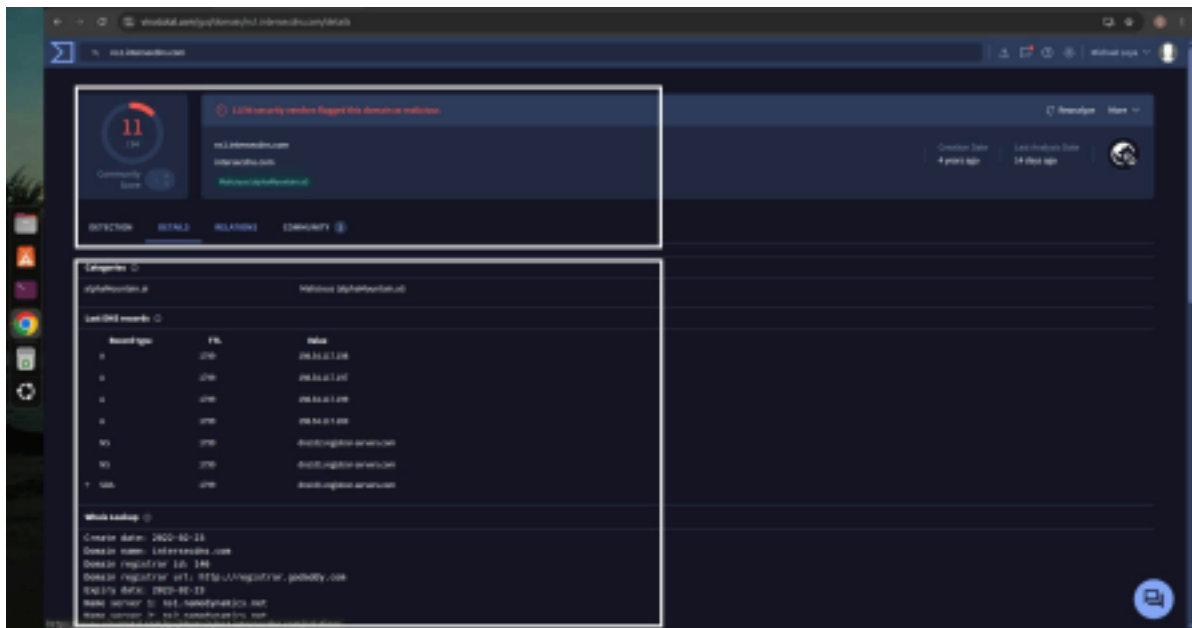


Figure 13: VirusTotal DNS & WHOIS Details – `ns1.intersecdns.com`

Forensic Breakdown:

- **Vendor Validation & Masquerading:** The domain is flagged as malicious by 11 security vendors. Similar to the **rootdnsservers** tactic, the actor utilises the string **intersecdns** to masquerade as legitimate internet security or sector-specific DNS infrastructure, aiming to bypass casual visual inspection by SOC analysts.
- **Registrar & Asset Diversification:** The WHOIS lookup reveals this domain was registered via a completely different provider (**GoDaddy**) in February 2022, contrasting with the 2019 Namecheap domains. This demonstrates high operational security (OPSEC). By diversifying their registrars and domain ageing, **Sea Turtle** ensures that a single vendor takedown or timeline-based blocklist cannot instantly cripple their entire hijacking operation.

4.7 Relational Infrastructure Mapping & Redundancy

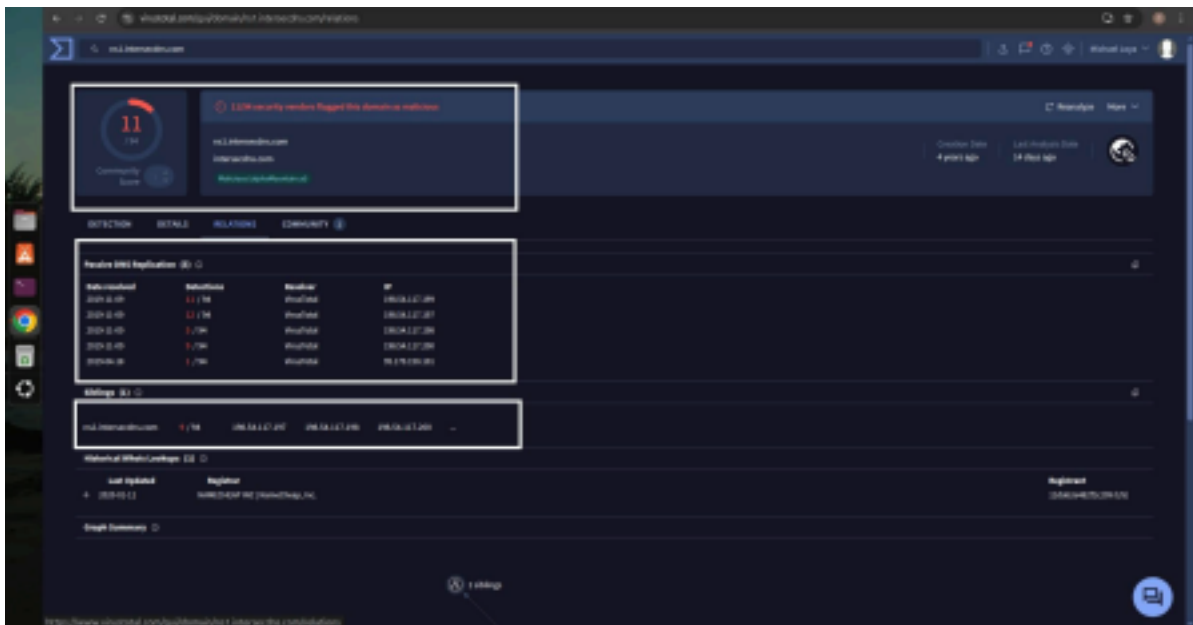


Figure 14: VirusTotal Relational Analysis & Passive DNS – `ns1.intersecdns.com`

Historical MISP Correlation: The Passive DNS Replication data provides critical validation of our initial threat feed. The telemetry shows that in **April 2019**, this domain resolved to **95.179.150.101**. This specific IP perfectly matches the Hosted malicious nameserver attribute extracted directly from **MISP Event #198**, mathematically confirming the accuracy of the historical intelligence.

4.8. MitM Infrastructure & Service Fingerprinting: To identify the operational characteristics of the adversary's Man-in-the-Middle (**MitM**) architecture, direct service fingerprinting was conducted using **Shodan** on the intersecdns campaign node **95.179.150.101**.

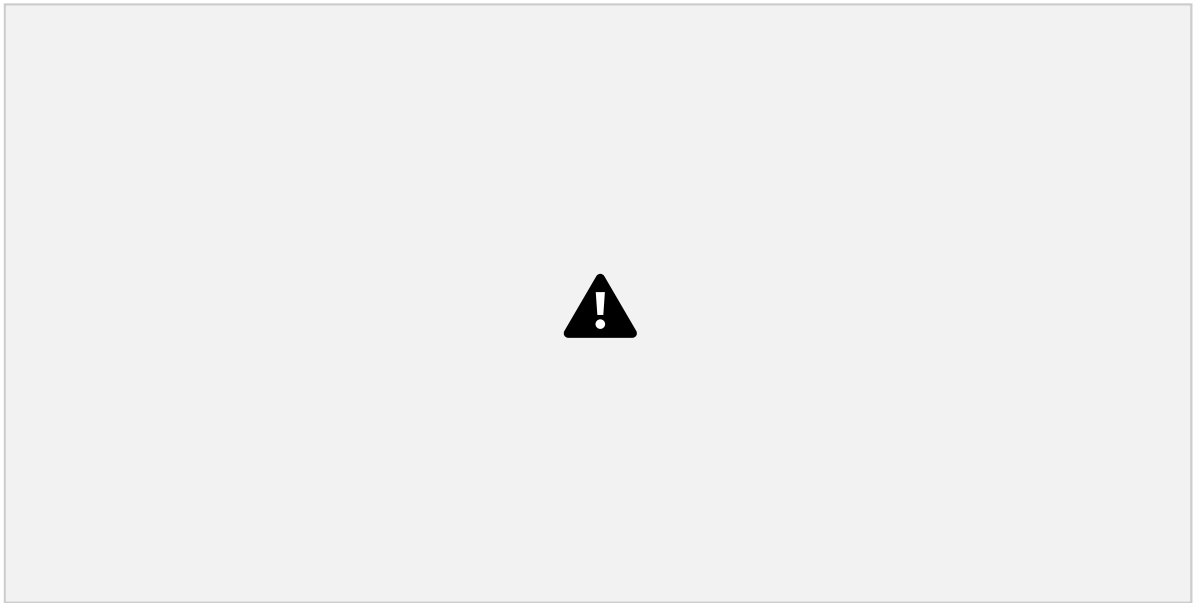


Figure 15: Shodan Fingerprinting - MitM Node 95.179.150.101

The most critical finding on this node is that it is actively listening on **Port 25 (SMTP)** and utilising the hostname **mail.vinoref.com**. The presence of an active mail server on a node utilised for DNS hijacking strongly indicates the adversary is specifically manipulating victim **MX (Mail Exchange)** records. This allows them to silently intercept, read, and forward corporate emails from compromised organisations without the victim ever realising their communications are compromised.

Geographic Blending: The server is hosted by **Vultr (The Constant Company, LLC)** and is physically located in **Haarlem, Netherlands**. By utilising cheap, disposable commodity VPS providers situated directly within their target geographic region, Sea Turtle effectively blends their malicious activity with normal, domestic Dutch network traffic.

MitM Infrastructure Fingerprinting: Encrypted Proxy Node Following the email node analysis, secondary service fingerprinting was conducted on the **rootdnservers** campaign node **95.179.156.61**, revealing a distinctly different operational purpose.

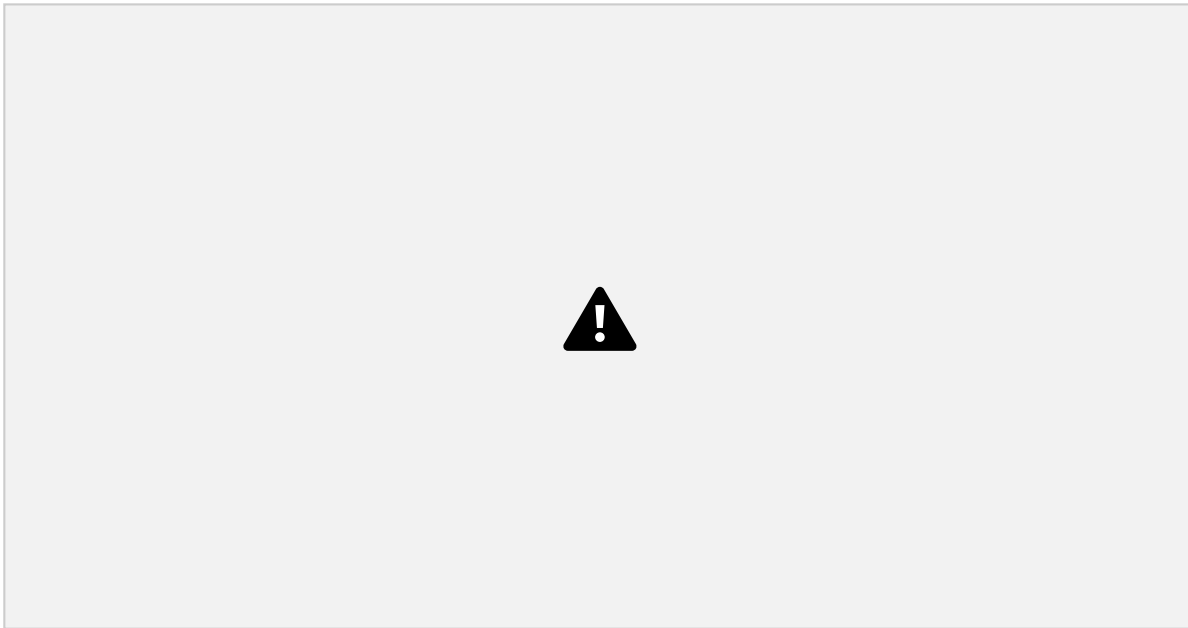


Figure 16: Shodan Fingerprinting – MitM Node 95.179.156.61

Encrypted Proxying (Port 443): This server is actively listening on **Port 443 (HTTPS)** and presents a valid **Sectigo** SSL certificate mapped to **.luminatinet.com**. Luminati (now **Bright Data**) is a well-known commercial proxy network. The threat actor is likely masquerading as or actively leveraging commercial proxy infrastructure to encrypt their intercepted web traffic. This effectively blinds defending SOCs, preventing them from performing Deep Packet Inspection (**DPI**) on the hijacked data.

Anomalous Services (Port 3337): The server also exposes an undocumented and highly anomalous service on **Port 3337**. In the context of an APT infrastructure node, non-standard high ports are frequently utilised as secondary administrative backdoors or dedicated staging channels for malware payload delivery.

5. Mitigation & Defence Strategy

Analysis of the **Sea Turtle APT** reveals a heavy reliance on bypassing perimeter firewalls by exploiting external DNS infrastructure and subsequently utilising valid administrative credentials. To neutralise these attack vectors, the following immediate mitigations are recommended:

5.1. DNS Layer Hardening (Preventing Initial Access)

Implement DNSSEC & Registry Lock: Mandate DNS Security Extensions (DNSSEC) to ensure DNS responses are cryptographically signed, preventing unauthorised redirection. Apply **Registry Lock** status to all tier-one corporate domains to require manual, out-of-band authorisation before any Nameserver (NS) records can be modified.

5.2. Administrative Portal Security (Neutralising cPanel Compromise)

- **Enforce Hardware MFA:** Standard TOTP (Authenticator apps) can be intercepted during a Man-in-the-Middle (**MitM**) DNS hijack. Transition all administrative accounts (cPanel, Domain Registrars, Cloud Consoles) to FIDO2 hardware security keys (e.g., YubiKey) to render intercepted passwords useless.
- **Strict Geo-Fencing & IP Allowlisting:** Since the adversary utilises European commercial VPNs to mask their logins, configure cPanel and administrative SSH access to strictly allowlist known, static corporate IP addresses, outright dropping all requests from commercial VPN and VPS subnets.

5.3. Linux Infrastructure Hardening (Restricting Lateral Movement)

- **Restrict SSH Pivoting:** Sea Turtle relies heavily on SSH to pivot through the network. Disable root SSH login globally. Enforce public-key (RSA/Ed25519) authentication exclusively, entirely disabling password-based SSH access. Implement strict rate-limiting on **Port 22** to detect and throttle brute-force or automated scripting attempts.

5.4. Immutable Logging (Defeating Anti-Forensics)

- To combat the adversary's TTP of wiping Linux system logs (**T1070.002**) and `.bash_history` (**T1070.003**), ensure all local logs are immediately forwarded to an off-site, immutable SIEM datastore. Configure syslog to transmit over TLS, ensuring that even if the host is compromised, the historical intrusion telemetry cannot be destroyed by the attacker.

6. Actionable Detection Logic (Splunk & Wazuh)

To operationalise the intelligence gathered in this report, the SOC Team must implement the following SIEM queries and EDR rules to actively hunt for SnappyTCP behaviour and Sea Turtle's anti-forensic footprint.

6.1. Splunk Detection: Anti-Forensic Execution (T1070)

Sea Turtle actively attempts to blind the SOC by destroying the command history. The SPL query below looks for explicit commands used to wipe Linux tracks.

Splunk SPL

```
index=linux sourcetype=bash_history OR sourcetype=linux_secure  
| search command IN ("rm *bash_history*", "history -c", "cat /dev/null >*", "ln -sf /dev/null  
~/.bash_history", "unset HISTFILE")  
| stats count min(_time) as firstTime max(_time) as lastTime by host, user,  
command | convert ctime(firstTime) ctime(lastTime)  
| rename host as Target_Host, user as Compromised_User
```

6.2. Splunk Detection: SnappyTCP Web Shell Activity (T1505.003) The adversary frequently utilises nohup (to keep the process running after logout) to spawn the SnappyTCP reverse shell. The query below detects suspicious background processing initiating external connections.

Splunk SPL

```
index=linux sourcetype=syslog OR sourcetype=linux_secure  
| search process="*nohup*" AND (process="*bash -i*" OR process="*socat*" OR  
process="*nc -e*")  
| eval Threat_Indicator="Suspected SnappyTCP Reverse Shell  
Execution" | table _time, host, user, process, Threat_Indicator
```

6.3. Wazuh EDR: File Integrity Monitoring for cPanel Web Directories: Sea Turtle drops the SnappyTCP malware and Adminer tools directly into public-facing web directories (like /var/www/html or /public_html). This Wazuh rule below ensures an immediate high-severity alert is fired if any .php, .sh, or ELF executable is dropped into a monitored web directory.

XML

<syscheck>

<directories check_all="yes" realtime="yes">/var/www/html</directories>

<directories check_all="yes" realtime="yes">/home/*/public_html</directories>

</syscheck>

<rule id="100050" level="12">

<if_sid>550</if_sid>

<match>\.php\$|\.sh\$</match>

<description>Sea Turtle APT Indicator: New script dropped in public web directory.</description>

<mitre>

<id>T1505.003</id>

</mitre>

<group>web_shell, apt_activity,</group>

</rule>